

Assembly Module

Current-state context, functionality and screen inventory

PLPI Batch Assembly Record workflow | Stage 11 | Updated 19 August 2026

ENTRY	OWNER	NEXT
Allocated assembly room	Assembly Team	Post-Assembly QC

1. Module context

The Assembly module is Stage 11 of the electronic Batch Assembly Record workflow. It receives batches after Production Control has completed verification and room allocation. Its controlled output is a completed, reconciled and electronically signed assembly record ready for Post-Assembly QC.

Workflow position: Pre-Assembly QC -> Production Control / Room Allocation -> Assembly Room -> Post-Assembly QC -> Pre-QP -> QP Release.

2. Business purpose

- Confirm that the correct batch, product, specimen and printed components are present in the allocated room.
- Record the batch start, team briefing and pre-start verification.
- Perform random sample and in-process checks during assembly.
- Reconcile issued, used, damaged and discrepant materials.
- Complete end-of-batch room clearance and retain an electronic audit trail.
- Move the finished batch to Post-Assembly QC only after all four pages are signed.

3. End-to-end user journey

Step	User action	System behaviour
1	Find batch	Search the active queue by B&S batch number or manufacturing lot.
2	Start batch	Confirm start; capture logged-in user and timestamp.
3	Initial checks	Verify BAR, specimen, allocated room and received printed materials.
4	Sample check	Confirm lot and expiry for each box received from Production Control.
5	IPC checks	Complete time-based in-process checks; add extra IPC rows if needed.
6	Reconcile	Record quantities received, used, damaged and discrepant; confirm room clearance.
7	Finish batch	Sign all four pages, record finish audit data and hand off to Post-Assembly QC.

4. Screen inventory

4.1 Assembly batch queue

Shows active and completed batches, batch/lot identifiers, description, quantity, allocated room, status and action. Search, refresh and queue tabs are interactive. The screen also presents check-in, check-out, break and partial-batch attendance controls with staff status rows.

Figure 1. Assembly Room workflow - active and completed batch queue

MachinePI Label PrintPPM AuditPPM LocationsPPM Audit HistoryRegulatory TaskRPM Pack CreationRPM ApprovalRegulatory LogProduct AddProductPacking ListGoods-In SummaryPre QIPer AssemblyBatch ChooserBMS Batch AddPrinter

Assembly Room

ASSEMBLY ROOM OPERATIONS

Assembly Batch Queue

View and manage active and completed assembly batches.

assembly.room

Assembly Room User

Active Batches

Completed Batches

Search by BMS Batch / Lot / Product

Search

Refresh

Active Batches

Batches allocated by Production Control and ready for assembly.

BMS BATCH NO.	MFG LOT NO.	DESCRIPTION	QUANTITY	ROOM	STATUS	ACTION
03A1582	25086A	Flutiform inhalation	650	Room 2	Not Started	Open
03A1582	25086A	Flutiform inhalation	250	Room 2	Not Started	Open
LVT914	MFG-GLIM-2414	Lumigan eye drops	390	Room 2	Not Started	Open
LVT915	MFG-AMIS-2415	Medical tablets	280	Room 3	Not Started	Open

Demonstration & Attendance

Record room attendance and partial batch activity.

Check In

Check Out

Break

Partial Start

Partial Finish

STAFF NAME	ROLE	CHECK IN	CHECK OUT	STATUS
JACINTA FER	Assembler	08:32	10:17	Checked out
SUTAI BEN	Assembler	08:33	10:19	Checked out
NASMEEN	Assembler	08:48	09:04	Present
MONICA CAR	Assembler	09:33	-	Checked out

Users

Leftoff Finding

Production Controller

Check IN/OUT

Assembly Rooms

Production Checking

QP Certified Batches

QP Release Dashboard

PUP Report

Logout

Exit

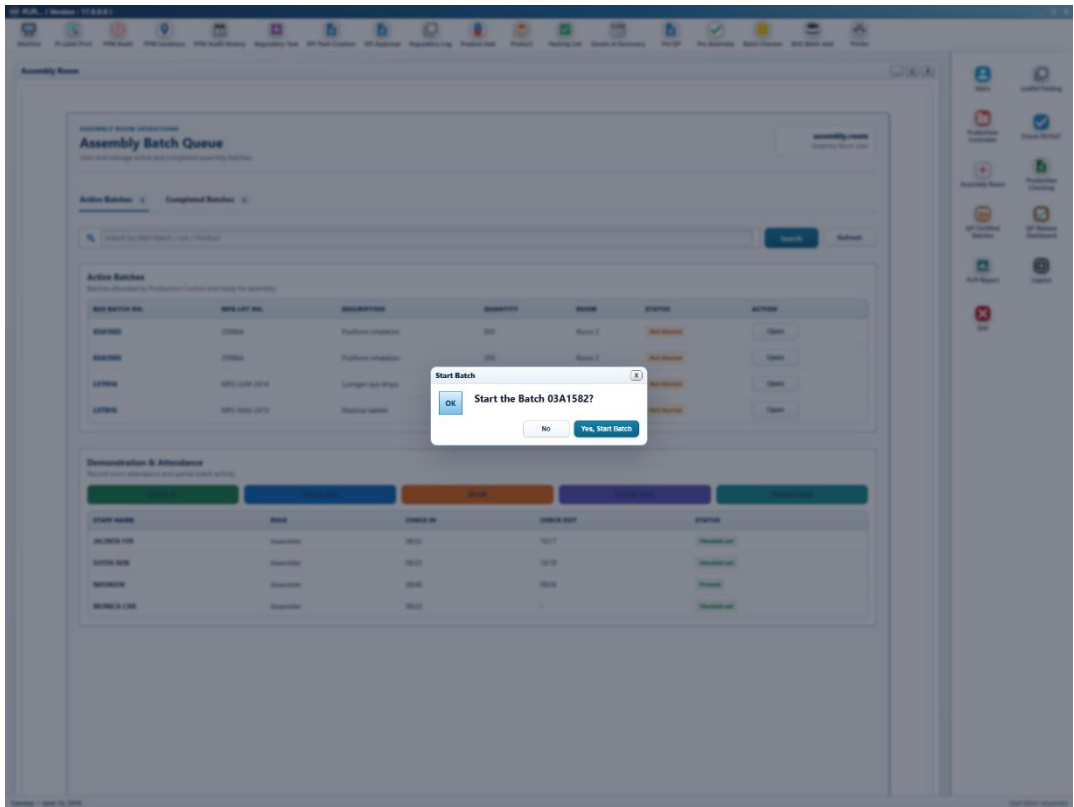
Tuesday -- June 16, 2021

03A1582 is ready in Assembly Room demo mode.

4.2 Start batch confirmation

Opening a batch that has not yet started raises a confirmation dialog. Accepting it records the current user and date/time as the Assembly Room batch start.

Figure 2. Start Batch confirmation and audit message



4.3 Initial Checks

Displays the batch identity strip, allocated room, lifecycle audit fields, briefing field and confirmations for the BAR, specimen and received materials. The page sign-off remains disabled until every required checkbox is selected.

Figure 3. Initial Checks and received-material verification

MachinePI Label PrintPPM AuditPPM LocationsPPM Audit HistoryRegulatory TaskRPA Pick CreationRPA ApprovalRegulatory LogProduct AddProductPacking ListGoods-In SummaryPre-QPPre AssemblyBatch CheckerBNS Batch AddPrinter

Assembly Room

Assembly Batch Guide

ASSEMBLY'S INITIAL CHECKS

Assembly Room - Initial Checks

Verify the batch, room, BAR, specimen and issued components before assembly begins.

BNS Batch No.
03A1582

Mfg Lot No.
25086A

Product
Flutiform Inhalation

Quantity
650

Allocated Room
Room 2

1 Initial Checks

2 Random Sample

3 IPC Checks

4 Recancellation

Lifecycle Audit

Batch start information captured automatically.

Batch Start
19 Aug 2026, 13:57:59

Started By
assembly:room

Assembly Room
Room 2

Team Briefing

Record the briefing or shift handover details.

Enter briefing details...

Initial Confirmations

All checks are required before Page 3 can be signed.

Batch Assembly Record (BAR) is available and correct

☐

Correct specimen (physical and electronic) is available

☐

Allocated room is clean and ready for assembly

☐

All required printed components received from Production Control

☐

All component boxes verified and accounted for

☐

Back to Queue

Complete all required checks to enable sign off
Initial Checks will be locked after signing

Sign Page 3

Users

Left-Hand Testing

Production Controller

Check IN/OUT

Assembly Rooms

Production Checking

QP Certified Satches

QP Release Dashboard

PPH Report

Logout

Exit

Tuesday - June 16, 2026

Batch 03A1582 started by assembly:room at 19 Aug 2026, 13:57:59

4.4 Random Sample Check

Creates one confirmation row per box passed from Production Control. Manufacturing lot and expiry are populated from the BAR details.

Figure 4. Random sample confirmation by box

PC PLP (Version: 17.0.0.0)

MachinePI Label PrintPPM AuditPPM LocationsPPM Audit HistoryRegulatory TaskRPN Pick CreationRPN ApprovalRegulatory LogProduct AddProductPicking ListGoods-In SummaryPre QPPre AssemblyBatch CheckerBNG Batch AddPrinter

Assembly Room

Assembly Batch Queue

Random Sample Check

Page 2 of 4

BNG Batch No.03A1582Mfg Lot No.25086AProductFlatlform InhalationQuantity650Allocated RoomRoom 2

Initial Checks

Random Sample

IPC Checks

Reconciliation

Box Confirmation

2 boxes carried from Production Control. Confirm each box against the BAR.

#	BOX NO.	MFG LOT NO.	EXPIRY DATE	STATUS	CONFIRMED BY	CONFIRMED AT
1	BOX 001	25086FA	07/2027	☐ Pending	-	-
2	BOX 002	25086FA	07/2027	☐ Pending	-	-

Back

Complete all required checks to enable sign off

Sign Page 2

Users

Locked Loading

Production Controller

Check IN/OUT

Assembly Room

Production Checking

QP Certified Batches

QP Release Dashboard

PLP Report

Logout

Exit

Tuesday - June 16, 2021

Batch 03A1582 started by assemblyroom at 19 Aug 2020, 15:57:58

4.5 IPC Checks

Displays estimated assembly time, minimum IPC count and scheduled checks. Operators can complete check rows, add extra IPC rows and attach photo evidence to the relevant check.

Figure 5. In-process checks and product identity strip

MachinePI Label PrintPPM AuditPPM LocationsPPM Audit HistoryRegulatory TaskRPV Pick CreationRPV ApprovalRegulatory LogProduct AddProductPacking ListGoods-In SummaryPre-QPPre AssemblyBatch CheckerBMS Batch AddPrinter

Assembly Room

Assembly Batch Guide

ANALYSIS > IN-PROCESS OPERATIONS

IPC Checks

Complete scheduled in-process checks and attach supporting photo evidence.

BMS Batch No.
03A1582

Mfg Lot No.
25086A

Product
Flutiform Inhalation

Quantity
650

Allocated Room
Room 2

1 Initial Checks

2 Random Sample

3 IPC Checks

4 Recalculation

IPC Checks

Complete every scheduled check and attach photo evidence where required.

Estimated Assembly Time
60 minutes

Minimum IPC Checks
3

Scheduled Checks
3

+ Add IPC Check

#	TARGET TIME	CHECK DESCRIPTION	RESULT	CHECKED BY	CHECKED AT	PHOTO EVIDENCE
1	00:20	Weight / Count Check	☐ Pending	-	-	Add photo
2	00:40	Appearance Check	☐ Pending	-	-	Add photo
3	00:60	In Process Check	☐ Pending	-	-	Add photo

Back

Complete all required checks to enable sign off
IPC Checks will be locked after signing

Sign Page 3

Users

Left-Handing

Production Controller

Check IN/OUT

Assembly Rooms

Production Checking

QP Certified Batchers

QP Release Dashboard

PPH Report

Login

Exit

Tuesday - June 16, 2021

Batch 03A1582 started by assembly.room at 15-Aug-2020, 15:57:55

4.6 Reconciliation and Closure

Captures component quantities received, used, damaged and discrepant. It also records total use, retention sample, yield, damages and comments, followed by end-of-batch room-clearance confirmations.

Figure 6. Material reconciliation and end-of-batch clearance

Assembly Room

Assembly Batch Queue
ASSEMBLY ROOM OPERATIONS

Reconciliation & Closure
Reconcile materials and complete end-of-batch room clearance.

Page 4 of 4
IN PROGRESS

BNS Batch No. 03A1582 Qty Lot No. 25086A Product Flatlume Inhalation Quantity 650 Allocated Room Room 2

1 Initial Checks 2 Random Sample 3 IPC Checks 4 Reconciliation

Material Reconciliation
Account for issued materials before completing room clearance.

ITEM	RECEIVED	USED	DAMAGED	DISCREPANT	BALANCE
Carton / Braille Label 1 FLU200-101200C2	657	650	0	7	0
Blister Label 1 FLU200-101200C2	657	650	0	7	0
Blister / Carton 2 FLU200-101200C2	657	650	0	7	0
Leaflet 1 FLU200-101200C2	657	650	0	7	0
Other Blank Label	657	650	0	7	0
Total Used	650				
Retention Sample		1			
Yield			649		
Total Damaged				0	

Comments
Record discrepancies, damage or relevant closure notes.
Enter comments (if any)...

End-of-Batch Room Clearance
All items are required before the batch can be finished.

- ☒ All materials returned to stores or properly accounted for
- ☒ Work area cleaned and equipment returned
- ☒ Waste disposed of as per SOP
- ☒ Room is clean and ready for the next batch

Back Complete all required checks to enable sign-off Finish & Sign All Pages

Tuesday - June 16, 2026 BATCH 03A1582 started by assemblyroom at 19 Aug 2026, 10:57:09

5. Functional requirements represented in the wireframe

Capability	Current behaviour	Maturity
Queue management	Interactive active/completed queues, search, refresh, room/status visibility	Interactive
Attendance	Check in/out, break and partial-batch activity feedback with staff status	Interactive prototype
Lifecycle audit	Start and finish user/date/time records	Interactive
Initial verification	BAR, specimen, room and component confirmation	Interactive with gating
Random samples	One confirmation per allocated box	Interactive with gating
IPC	Calculated minimum checks, additional rows and photo evidence capture	Interactive
Reconciliation	Received, used, damages, discrepancy, yield and comments	Interactive; discrepancy auto-calculated
Clearance	Four end-of-batch clearance checks	Interactive with gating
Electronic sign-off	User/timestamp per page; page locks after sign-off	Interactive prototype
Handoff	Moves completed batch to Post-Assembly QC	Interactive prototype

6. Rules and completion gates

- The Initial Checks page cannot be signed until all material and pre-start confirmations are complete.
- The Random Sample Check page cannot be signed until every box is confirmed.
- The IPC page cannot be signed until all displayed IPC checks are complete.

- Reconciliation cannot be signed until its required checks and quantity fields are complete.
- Each signed page is locked and displays the signing user and timestamp.
- Signing the fourth page triggers Finish Batch confirmation and the Post-Assembly QC handoff.

7. Current prototype gaps to address in redesign

- Barcode scanning is displayed but has no scan validation or error-handling workflow.
- There is no deviation, investigation or supervisor-override workflow for a non-zero discrepancy.
- Electronic sign-off is represented by a confirmation dialog rather than validated re-authentication.
- Prototype records are in memory and do not demonstrate durable storage, versioning or audit export.
- Attendance controls provide prototype feedback but are not connected to a persistent workforce or timekeeping service.
- The reconciliation table pre-populates repeated example quantities across material types and needs component-specific data binding.

8. AI redesign brief

Redesign the Assembly module as a tablet-friendly, GxP-aware workflow for operators working in an allocated room. Preserve traceability and completion gates while reducing visual density and making the current task, outstanding requirements and exceptions immediately visible.

Essential redesign outcomes

- A clear progress stepper for Initial Checks, Samples, IPC and Reconciliation.
- Persistent batch identity, room, quantity, status and active operator context.
- Scan-first batch/component verification with visible match or mismatch feedback.
- Large touch targets and a focused task view suitable for tablets and handheld devices.
- Component-specific reconciliation with live expected balance and prominent discrepancy status.
- Evidence capture attached to the relevant IPC, exception or component row.
- A dedicated exception path with reason, evidence, supervisor review and resolution status.
- Validated electronic signature, audit history and clear locked/read-only states.
- Offline/interruption recovery, draft state and safe resume behaviour.
- Accessible colour contrast with status communicated by text and icon as well as colour.

Suggested prompt for a design AI

Using the supplied Assembly Module reference document and screenshots, redesign the Stage 11 Assembly Room workflow for a regulated pharmaceutical batch-record system. Create a desktop and tablet design with an active-batch queue, scan-first batch start, four-step guided checklist, component-level reconciliation, IPC evidence capture, exception handling, validated electronic signatures and a complete audit trail. Preserve all displayed batch identifiers and completion gates, but simplify hierarchy, reduce dense tables and make incomplete, warning, discrepancy, signed and locked states unambiguous. Produce screen variants for normal flow, mismatch, non-zero discrepancy, evidence upload, signature confirmation and completed handoff.